

and transmits information across. This router could be a part of a network of routers which then connects to the Internet as well, further broadening the reach of the network. This is the typical setup for network connections even today across most spaces such as offices and home environments.

The information traffic across a network is done in small chunks known as packets. For example when we tell the computer to print a document on our wireless network, the document data is split up into smaller parts and sent to the printer. The printer then receives all these small parts and reassembles them to print out the document. This system allows for the stability and security of the data, since if files were sent in one big chunk then any



The IoT network is expected to extend invisibly across the globe.

disruption in the network would cause the data to be corrupted en route and also slow down the transmission speeds. This breaking down of data into packets allows large quantities of data to be transmitted seamlessly which also prevents the connection pipeline from becoming jammed or overflowing. The breaking and rebuilding of the data is a highly precise process which requires hardware compatibility between the devices.

The means through which the networked hardware interact to accomplish their task is through the means of a network transfer protocol. These protocols or rules define how data is managed and treated across the whole network. The standard that is currently used on almost all networks, including the Internet and the Web, is known as Transmission Control Protocol/Internet Protocol or TCP/IP. The Internet Protocol is the component of this